

Title: Explainable AI Framework for Spam SMS Detection Using Transformers

Abstract

Spam SMS messages have become a major challenge in modern communication systems due to their increasing use in phishing attacks, financial fraud, malware distribution, and unsolicited advertisements. Traditional machine learning techniques such as **Naïve Bayes**, **Logistic Regression**, and **Support Vector Machine (SVM)** have been widely used for spam detection. However, these methods often rely on manually engineered features and fail to capture the contextual meaning of text messages, leading to lower prediction accuracy. Moreover, they provide little or no explanation for their decisions, making it difficult for users to trust the classification results.

The proposed project, "Explainable AI Framework for Spam SMS Detection Using Transformers," aims to develop an intelligent and transparent spam detection system using **Bidirectional Encoder Representations from Transformers (BERT)**. The system utilizes the SMS Spam Collection Dataset to fine-tune a pre-trained BERT model capable of understanding the semantic and contextual information present in SMS messages. After preprocessing the input text through cleaning, tokenization, and encoding, the model classifies each message as either **Spam or Ham (Non-Spam)** with high accuracy.

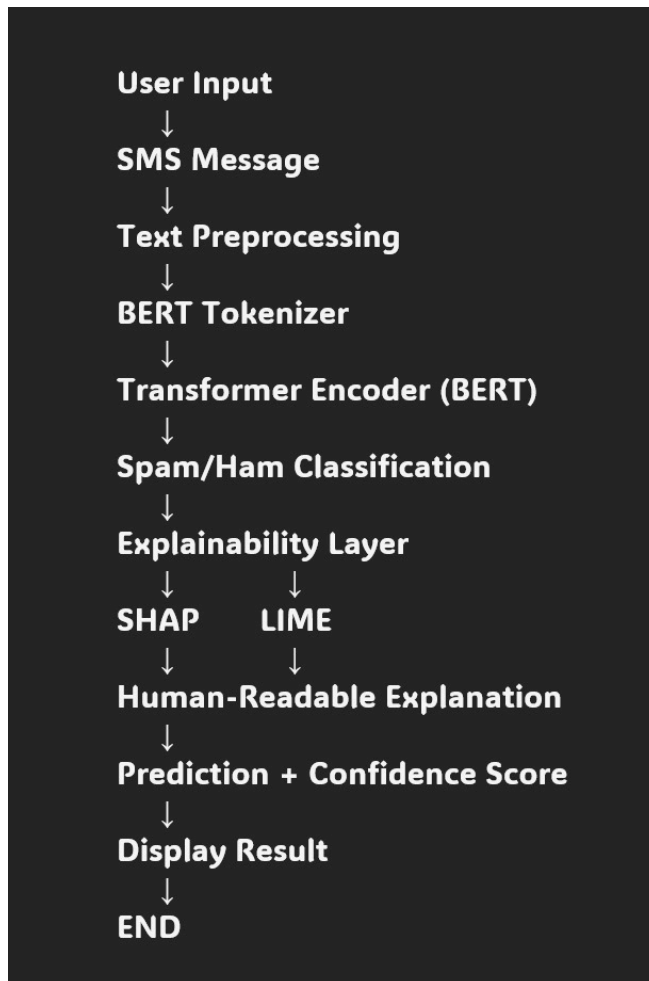
To enhance the transparency and interpretability of the model, the proposed framework integrates **Explainable Artificial Intelligence (XAI)** techniques such as **SHAP (SHapley Additive exPlanations)** and **LIME (Local Interpretable Model-Agnostic Explanations)**. These techniques identify and highlight the important words or phrases that contribute most to the model's prediction, enabling users to understand why a particular SMS is classified as spam or legitimate.

The system is implemented as a simple web application that allows users to enter SMS messages and instantly receive the classification result, confidence score, and a human-readable explanation. The performance of the proposed model is evaluated using standard metrics such as **Accuracy, Precision, Recall, and F1-Score**, and compared with traditional machine learning algorithms. The proposed framework aims to provide an accurate, reliable, and explainable spam detection solution that enhances user trust while effectively protecting users from unwanted and malicious SMS messages.

Datasets

Dataset	URL link
SMS Spam Collection Dataset	https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset
SMS Spam Collection(Hugging Face)	https://huggingface.co/datasets/Deysi/spam-detection-dataset
SMS Spam Collection(Hugging Face)	https://huggingface.co/datasets/ucirvine/sms_spam

Approach



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